

One aperture, two geometric outcomes

One fixed aperture admits disk A and blocks disk B solely by their diameters (Fig. 1). In this ideal two-dimensional DEMO, the opening width is 3, while the diameters are 2 and 4. Passage requires $d < 3$; even equality is blocked.

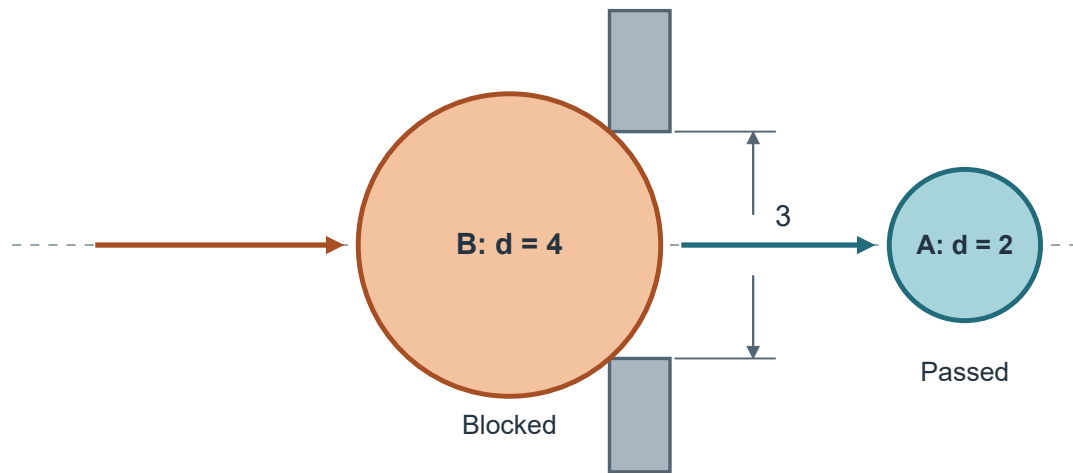


Figure 1. Geometric outcomes at the same aperture (constructed DEMO). Disk diameters and opening width share one scale; lengths are in arbitrary units. A is placed downstream to show passage, while B contacts the opening edges and remains upstream. Arrows indicate left-to-right approach and passage, not measured trajectories. The rule is strict: $d = 3$ is blocked.

Both rigid disks approach the rigid barrier from left to right along the opening centerline. Constructed checks at $d = 2, 2.5, 3$, and 4 return pass, pass, block, and block; these cover the threshold and its boundary, not statistical variation. The model omits elasticity, friction, adhesion, forces, and clogging dynamics. It supplies no efficiency, throughput, or probability evidence and proposes no new sieving law.